MACHINE LEARNING AND DEEP LEARNING

Lab 06: Semi-Supervised Learning

Dr. Giacomo Lucato, a.y. 2025-2026

SEMI-SUPERVISED LEARNING

- **Supervised Learning** is a Machine/Deep Learning family of tasks in which each training sample comes with a **label**, telling the category, or class, to which the sample belongs. The goal is to train a model able to extract **similar patterns** from data belonging to the same **given class**, so that the model will be able to assign each unseen data sample the correct class.
- **Unsupervised Learning** is the opposite: samples do not come with a label indicating the ground truth class, therefore the goal becomes teaching a model how to **group** together samples with similar structure. In other words, the model should be able to extract patterns relevant to group data into **different clusters,** where elements from the same cluster should be similar, while elements from different clusters should be dissimilar.
- **Semi-Supervised Learning (SSL)** stays in the middle: the dataset is composed of **few labeled samples** and **many unlabeled samples.** To solve SSL, it is necessary to combine strategies coming both from Supervised and Unsupervised Learning.

Π-MODEL

- **Π-Model** is based on the concept of **consistency regularization:** if the same input, with some slight stochastic changes, is given twice, the model should return similar outputs.
- This pushes the model to extract feature representations solid enough to be **invariant** to small perturbations. The loss is given by the **difference** between the two representations of the two variants of the same input.
- In practice, this behavior is obtained by adding **Dropout:** the model is equipped with a Dropout Layer, that randomly deactivates a percentage of neurons every time an input is given.
- This ensures that every time the same input is given to the model the **internal representation changes slightly,** because the active neurons are different every time.

Π-MODEL

- The few labeled samples are then used to perform **standard classification,** through the computation of the usual Cross-Entropy Loss.
- The final loss the model must minimize is given by a weighted sum between the classification loss and the representation (consistency) loss:

$$Loss = \text{Supervised Loss} + \lambda(t) \cdot \text{Consistency Loss}$$

- The term λ ensures that the consistency loss is **not suffocated** by the supervised loss, by assigning more weight to it.

DEEP CLUSTERING

- **Deep Clustering** is a model typically used to solve Unsupervised Learning but can be easily extended to Semi-Supervised Learning.
- In standard Unsupervised Learning, the goal is to **group** samples in clusters based on similarities between the samples. The model should learn how to extract a valid representation from each sample to assign it to the **correct cluster.**
- To do so, Deep Clustering relies on a technique known as **self-teaching:** when an unlabeled samples is given, the model outputs a **probability distribution** $\boldsymbol{q}$ spreading across all clusters, telling for each cluster the probability of the sample belonging to it. We refer to this distribution as **soft guess.**

DEEP CLUSTERING

- This probability q is then **sharpened,** to make the guessing less uncertain. This creates the **target distribution p.**
- The loss can be then computed as the difference between q and p. This is done through the **Kullback-Leibler Divergence:**

$$KL(P||Q) = \sum P(i) \log \frac{P(i)}{Q(i)}$$

- This metric measures the **distance** between two probabilities distributions or, more precisely, how well q **approximates** the target probability p.
- Intuitively, if p and q are the same, their ratio simplifies to 1 and the logarithm, with the all loss, collapses to 0.

DEEP CLUSTERING

- The output probability q is measured by computing the **distance** between the input sample and the **center** of each possible cluster, expressed as coordinates in the features' space.
- These centers are trainable **parameters** of the model, that need to be adjusted minimizing the loss between q and p.
- Instead of initializing them randomly, we run the **KMeans algorithm** over the training set, using the resulting clusters' centroids. This speeds up computations and increases precision.

DEEP CLUSTERING

- The few labeled samples are used to perform **standard classification**, computing Cross-Entropy loss between model's prediction and ground truth label.
- The final loss is given by a **weighted sum** between Cross-Entropy and KL Divergence.

EXERCISE

- The goal of the exercise is to construct a **Π-Model** and a **Deep Clustering** model, training and evaluating them over the MNIST dataset, modified to have few labeled samples and many unlabeled ones.
- **Task:** download and complete *Semi_Supervised_Learning.ipynb*, following the instructions provided in the comments.